UNIT 5

Knowledge representation

- Knowledge: Facts, information, and skills acquired through experience or education
- Knowledge Representation: Representing information about the real world in some formal language.
- It also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

Techniques used for Knowledge Representation

- Propositional Logic
- First Order Logic
- Rule Based System
- Semantic Networks
- Frames
- Script

Propositional logic

- Represent the facts, which are either true or false.
- Example: a)There are 7 days in a week(True)

  b) The Sun rises from West (False)

  c) 3+3= 7(False)
- Propositional logic is also called Boolean logic as it works on 0 and 1.
- Statements which are questions, commands, or opinions are not propositions such as **"How are you"**, **"What is your name"**, are not propositions.
- Symbols: A, B, C, P, Q, R, etc.
- Connectives: ^ , ˅ , ˜ , → , ⇔

Propositional Logic Connectives

Terms	Connective Symbol	Meaning
Conjunction	$\wedge$	AND
Disjunction	$\vee$	OR
Negation	$\sim$	NOT
Conditional	$\rightarrow$	IF..THEN
Biconditional	$\Leftrightarrow$	IF AND ONLY IF

Translation of English sentences to propositional logic

Peter can play tennis	P
Peter cannot play tennis	$\sim P$
Peter can play tennis and badminton	$P \wedge Q$
Peter can play tennis or badminton	$P \vee Q$
If Peter can play tennis then he can play badminton	$P \rightarrow Q$
Peter can play tennis if and only if he can play badminton	$P \leftrightarrow Q$

Truth Table

P	Q	$P \wedge Q$	$P \vee Q$	$P \rightarrow Q$	$P \leftrightarrow Q$	~P
T	T	T	T	T	T	F
T	F	F	T	F	F	F
F	T	F	T	T	F	T
F	F	F	F	T	T	T

First Order Logic

- In propositional logic, we can only represent the facts, which are either true or false. e.g. “It is raining”
- But Propositional logic is not sufficiently expressive to represent the following statements

  "Some humans are intelligent"

  “All mangoes are sweet“
- First-order logic is capable of expressing facts about some or all objects in the universe.
- FOL specifies objects, relations and functions.
- First-order logic is also known as **Predicate logic.**

First Order Logic

- Syntax: function (term1,term 2,...,term n)

  Predicate (term1, term2, ..., term n)

X is a girl → **girl(x)**

Subject

Predicate

Quantifiers

Quantifiers

Universal Quantifier → ∀ -For all → All employees → $\forall x\ emp(x)$

Existential Quantifier → ∃ -For some → Some students → $\exists x\ student(x)$

Elements of FOL

Constant	1, 2, A, John, Mumbai, cat
Variables	x, y, z, a, b, etc.,
Predicates	Greater(5,3), Student(boy, girl).
Function	Mother-of(John) , Colour-of (Basket)
Connectives	$\wedge$, $\vee$, $\neg$, $\Rightarrow$, $\Leftrightarrow$
Equality	==
Quantifier	$\forall$,(universal quantifier) $\exists$,(existential quantifier)

FOL-Example

- **Sam is tall**

  tall(Sam)
- **John likes cricket.**

  likes(John, cricket).
- **Everyone likes cricket.**

  $\forall x$ likes(x, cricket)
- **All students like cricket.**

  $\forall x$ (student(x) $\rightarrow$ like(x, cricket))
- **Some boys are intelligent**

  $\exists x$(boys(x) $\wedge$ intelligent(x))
- **Some integers are even and some are odd**

  $\exists x$ even(x) $\wedge$ $\exists x$ odd(x)
- **Not all students like both Mathematics and Science.**

  $\neg\forall$ (x) (student(x) $\rightarrow$ like(x, Mathematics) $\wedge$ like(x, Science))

$\wedge$	AND
$\vee$	OR
$\sim$	NOT
$\rightarrow$	IF..THEN
$\Leftrightarrow$	IF AND ONLY IF

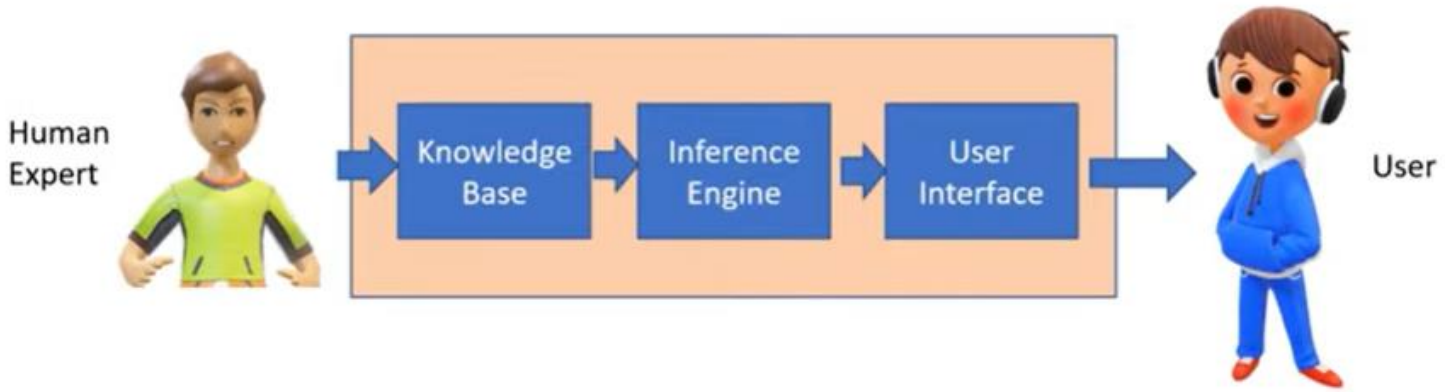
$\forall$	$\rightarrow$
$\exists$	$\wedge$

Forward chaining and backward chaining

Inference engine

- It applies logical rules to the knowledge base to infer new information from known facts.
- Inference engine commonly proceeds in two modes:

  1.Forward chaining 2.Backward chaining



Forward Chaining

- Forward chaining starts with the available data and uses inference rules to extract more data until a goal is reached.
- **"What will happen next?"**
- It is a bottom up approach.
- Forward chaining is data driven because the reasoning starts from a set of data.

Forward Chaining-Example

- Example

  It is raining.
  If it is raining, the street is wet.
  The street is wet.

- Conclude from "A" and "A implies B" to "B“

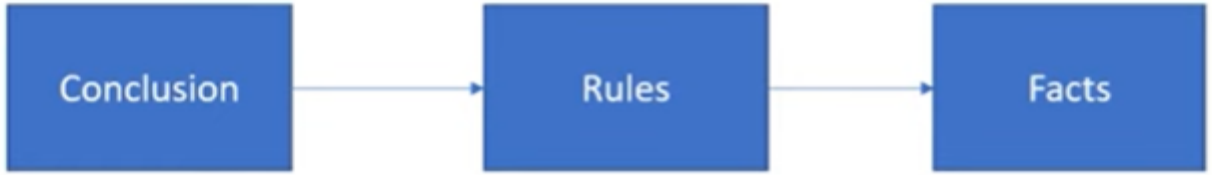
  A

  A->B

  B

Backward Chaining

- Backward chaining starts from goal and proceeds backward to determine the set of rules that match the goal.
- **"Why did this happen?"**
- It is top down approach.
- Forward chaining is goal driven
- Example: diagnose bacterial infections.



Backward Chaining- Example

- Example:

  The street is wet.
  If it is raining, the street is wet.
  It is raining.

- Conclude from "B" and "A implies B" to "A".

  B

  A->B

  A

Example

- The law says that it is a crime for an American to sell weapons to hostile nations. Country Q, an enemy of America, has some missiles, and all of its missiles were sold to it by Jack, who is American.
- Prove that **"Jack is criminal".**

- it is a crime for an American to sell weapons to hostile nations

  $American(x) \land Weapon(y) \land Sells(x,y,z) \land Hostile(z) \Rightarrow Criminal(x)$
- Q has some missiles

  $\exists x\ Owns(Q,x) \land Missile(x)$

  $Owns(Q,M1)$

  $Missile(M1)$

1. Convert it into a simple sentence
2. Convert to FOL

Conversion into FOL

- all of its missiles were sold to it by Jack

  $\forall x\ Missile(x) \wedge Owns(Q,x) \Rightarrow Sells(Jack,x,Q)$

  $Missile(x) \wedge Owns(Q,x) \Rightarrow Sells(Jack,x,Q)$
- Missiles are weapons

  $Missile(x) \Rightarrow Weapon(x)$
- Enemies of America are hostile

  $Enemy(x, America) \Rightarrow Hostile(Q)$
- Jack, who is American

  $American(Jack)$
- The country Q, an enemy of America

  $Enemy(Q,America)$

Forward Chaining->Facts,Rules,Conclusion

1. American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)
2. Owns(Q,M1)
3. Missile(M1)
4. Missile(x) ∧ Owns(Q,x) ⇒ Sells(Jack,x,Q)
5. Missile(x) ⇒ Weapon(x)
6. Enemy(x, America) ⇒ Hostile(Q)
7. American(Jack)
8. Enemy(Q,America)

Step 1-Forward Chaining

Owns(Q,M1)

Missile(M1) => **Known Facts**

American(Jack)

Enemy(Q, America)

American(Jack) | Missile(M1) | Owns(A,M1) | Enemy(A, America)

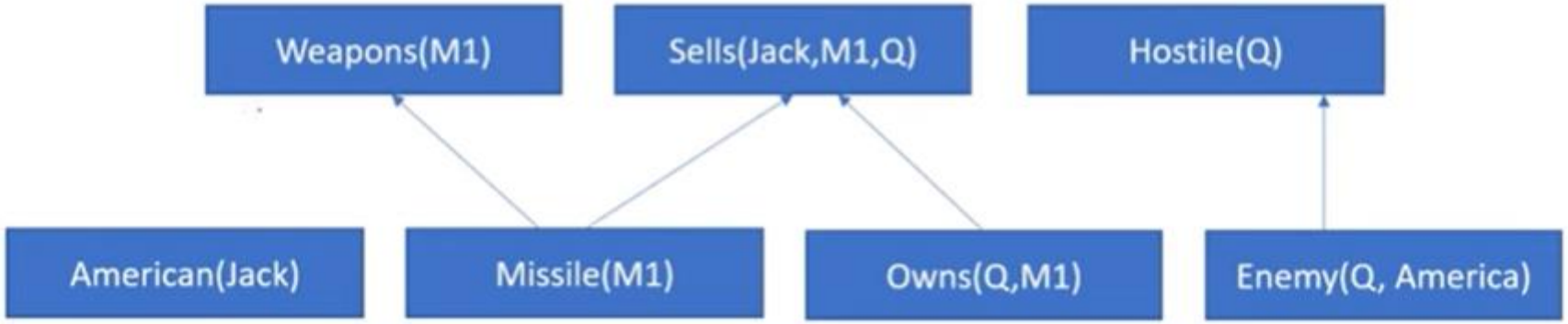
Forward Chaining->Facts,Rules,Conclusion

1. American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)
2. Owns(Q,M1)
3. Missile(M1)
4. Missile(x) ∧ Owns(Q,x) ⇒ Sells(Jack,x,Q)
5. Missile(x) ⇒ Weapon(x)
6. Enemy(x, America) ⇒ Hostile(Q)
7. American(Jack)
8. Enemy(Q,America)

Step 2-Forward Chaining

- Missile(x) ⇒ Weapon(x)
- ∀x Missile(x) ∧ Owns(Q,x) ⇒ Sells(Jack,x,Q)
- Enemy(x, America) ⇒ Hostile(Q)

=> **Rules that satisfy the facts**



Forward Chaining->Facts,Rules,Conclusion

1. American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)
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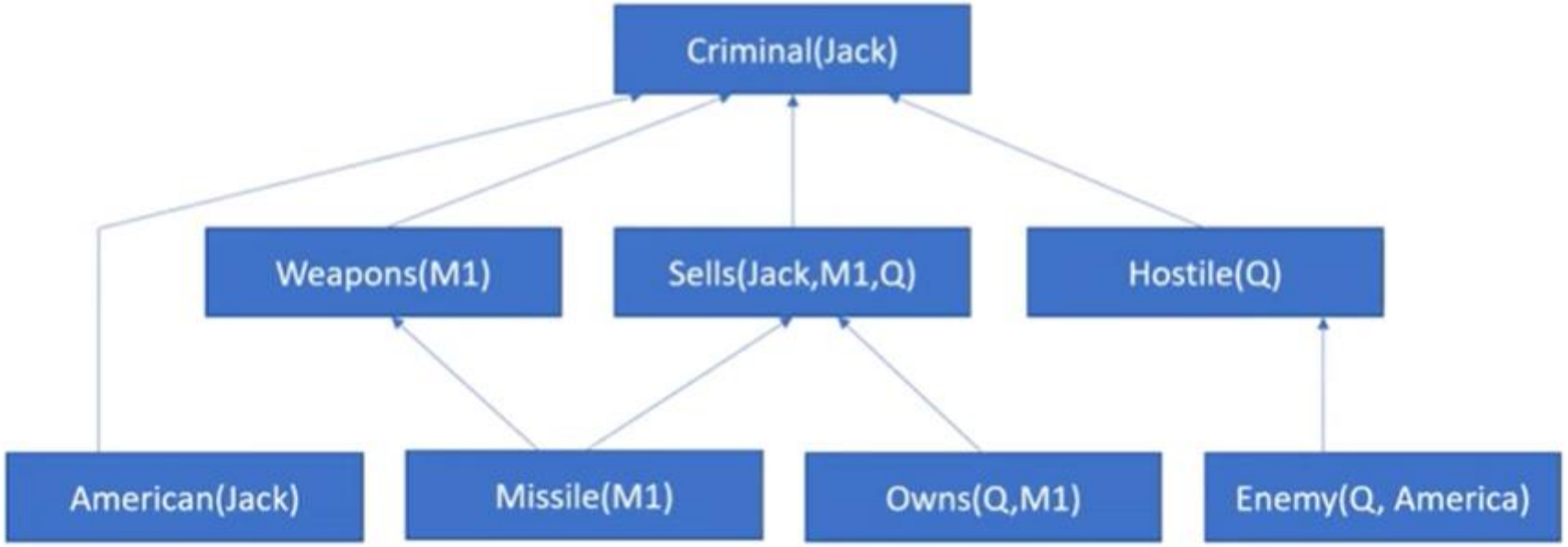
4. Missile(x) ∧ Owns(Q,x) ⇒ Sells(Jack,x,Q)
5. Missile(x) ⇒ Weapon(x)
6. Enemy(x, America) ⇒ Hostile(Q)
7. American(Jack)
8. Enemy(Q,America)

Step 3-Forward Chaining

- American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)

Proof



Backward Chaining
Conclusion,Rules,Facts

1. American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)
2. Owns(Q,M1)
3. Missile(M1)
4. Missile(x) ∧ Owns(Q,x) ⇒ Sells(Jack,x,Q)
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7. American(Jack)
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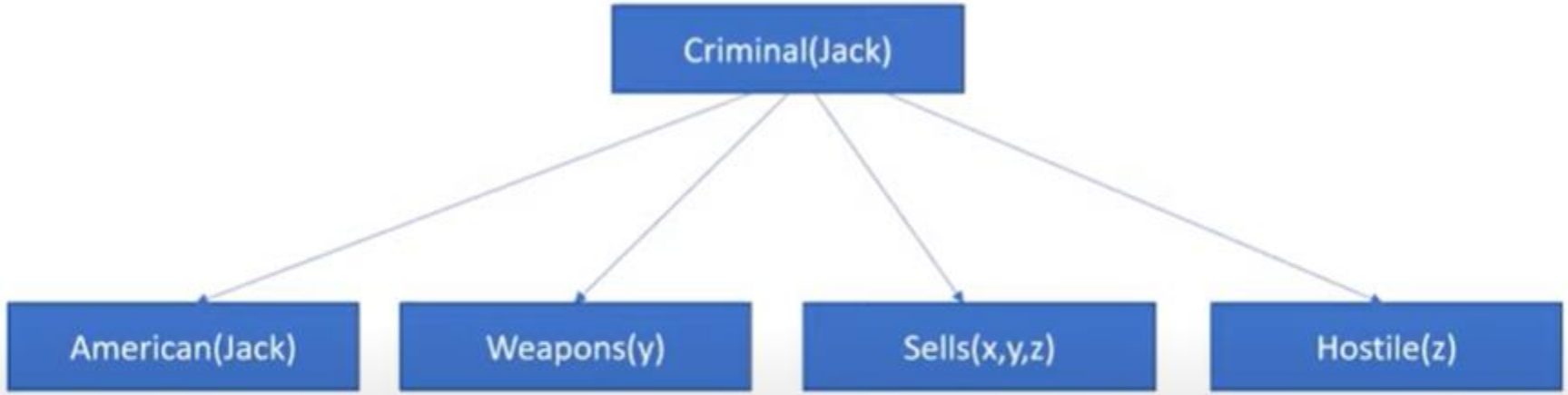
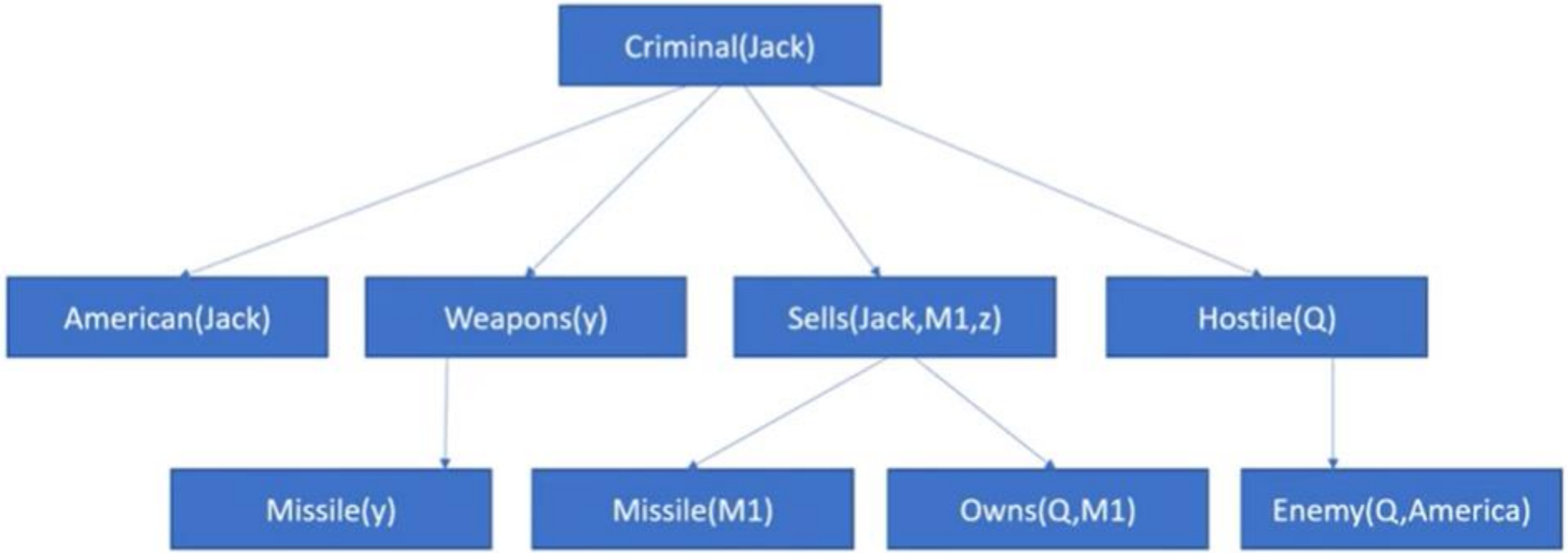
Backward Chaining-Conclusion,Rules,Facts



Backward Chaining-Conclusion,Rules,Facts



Module -5 Knowledge Representation and Reasoning for various AI applications Statistical Reasoning:

- Probability and Bayes' Theorem, Certainty Factors and Rule-Base Systems, Bayesian Networks, Dempster-Shafer Theory, Fuzzy Logic.
- AI for knowledge representation, rule-based knowledge representation, procedural and declarative knowledge, Logic programming, Forward and backward reasoning.
- Applications: Expert Systems, Decision Support System, Generative AI in Natural Language Processing (NLP) AI-based Programming tools.

Uncertainty and reasoning

- With this knowledge representation, we might write A→B, which means if A is true then B is true, but consider a situation where we are not sure about whether A is true or not then we cannot express this statement, this situation is called uncertainty.
- So to represent uncertain knowledge, where we are not sure about the predicates, we need uncertain reasoning or probabilistic reasoning

Causes of uncertainty:

Following are some leading causes of uncertainty to occur in the real world.

1. Information occurred from unreliable sources.
2. Experimental Errors
3. Equipment fault
4. Temperature variation
5. Climate change.

Probabilistic reasoning

Need of probabilistic reasoning in AI:

- When there are unpredictable outcomes.
- When specifications or possibilities of predicates becomes too large to handle.
- When an unknown error occurs during an experiment.

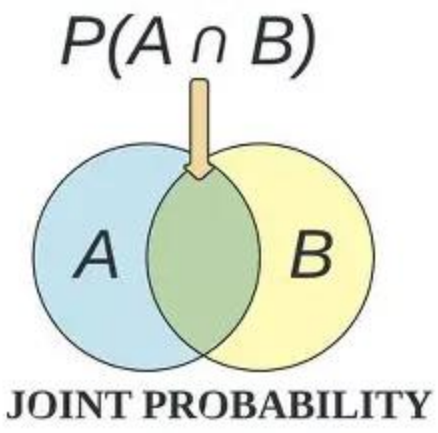
In probabilistic reasoning, there are two ways to solve problems with uncertain knowledge:

- Bayes' rule
- Bayesian Statistic

Probabilistic reasoning

Joint Probability

- Joint probability is the likelihood of **more than one event** occurring at the **same time** P(A and B).
- The probability of event A and event B occurring together. It is the probability of the intersection of two or more events written as **p(A ∩ B)**.



Conditional Probability

- The conditional probability of an **event *B*** is the probability that the event will occur given the knowledge that an **event *A* has already occurred**. It is denoted by **P(B|A).**
- So now, The joint probability of two dependent events becomes
- **P(A and B) = P(A)P(B|A)**

Bayes' theorem (Conditional Probability)

Bayes' theorem can be derived using product rule and conditional probability of event A with known event B:

As from product rule we can write:

> $P(A \wedge B) = P(A|B)\, P(B)$ or

Similarly, the probability of event B with known event A:

> $P(A \wedge B) = P(B|A)\, P(A)$

Equating right hand side of both the equations, we will get:

$$P(A|B) = \frac{P(B|A)\, P(A)}{P(B)} \quad \text{....(a)}$$

The above equation (a) is called as **Bayes' rule** or **Bayes' theorem**. This equation is basic of most modern AI systems for **probabilistic inference**.

Bayes' theorem (Conditional Probability)

It shows the simple relationship between joint and conditional probabilities. Here,

P(A|B) is known as **posterior**, which we need to calculate, and it will be read as Probability of hypothesis A when we have occurred an evidence B.

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

P(B|A) is called the likelihood, in which we consider that hypothesis is true, then we calculate the probability of evidence.

P(A) is called the **prior probability**, probability of hypothesis before considering the evidence

P(B) is called **marginal probability**, pure probability of an evidence.

In the equation (a), in general, we can write P (B) = P(A)*P(B|Ai), hence the Bayes' rule can be written as:

$$P(A_i|B) = \frac{P(A_i)*P(B|A_i)}{\sum_{i=1}^{k} P(A_i)*P(B|A_i)}$$

Where A_1, A_2, A_3,....., A_n is a set of mutually exclusive and exhaustive events.

Bayes' theorem Example

What is the probability that a patient has diseases meningitis with a stiff neck? A doctor is aware that disease meningitis causes a patient to have a stiff neck, and it occurs 80% of the time. He is also aware of some more facts, which are given as follows:

- The Known probability that a patient has meningitis disease is 1/30,000.
- The Known probability that a patient has a stiff neck is 2%

Soln

P(a|b) = 0.8

P(b) = 1/30000

P(a)= .02

$$\mathbf{P(b|a)} = \frac{P(a|b)P(b)}{P(a)} = \frac{0.8*(\frac{1}{30000})}{0.02} = 0.001333333.$$

Bayes' theorem Example

Consider the following data. Depending on the weather (sunny, rainy or overcast), the children will play(Y) or not play(N).

	Outlook	Play
1	Rainy	Yes
2	Sunny	Yes
3	Overcast	Yes
4	Overcast	Yes
5	Sunny	No
6	Rainy	Yes
7	Sunny	Yes
8	Overcast	Yes
9	Rainy	No
10	Sunny	No
11	Sunny	Yes
12	Rainy	No
13	Overcast	Yes
14	Overcast	Yes

Frequency table for the Weather Conditions

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	5

Bayes' theorem Example

Frequency table for the Weather Conditions

Weather	Yes	No
Overcast	5	0
Rainy	2	2
Sunny	3	2
Total	10	~~5~~ 4

Likelihood table weather condition

Weather	No	Yes	
Overcast	0	5	5/14= 0.35
Rainy	2	2	4/14=0.29
Sunny	2	3	5/14=0.35
All	4/14=0.29	10/14=0.71	P(yes)

Bayes' theorem Example

If the weather is sunny, then the Player should play or not?

P(Yes|Sunny)= P(Sunny|Yes)*P(Yes)/P(Sunny)

P(Sunny|Yes)= 3/10= 0.3

P(Sunny)= 0.35

P(Yes)=0.71

So P(Yes|Sunny) = 0.3*0.71/0.35= **0.60**

P(No|Sunny)= P(Sunny|No)*P(No)/P(Sunny)

P(Sunny|NO)= 2/4=0.5

P(No)= 0.29

P(Sunny)= 0.35

So P(No|Sunny)= 0.5*0.29/0.35 = **0.41**

So as we can see from the above calculation that **P(Yes|Sunny)>P(No|Sunny)**

Hence on a Sunny day, Player can play the game.

Dempster Shafter Theory

- **The Dempster -**
  Shafer theory (DST) is a mathematical framework for reasoning with uncertainty, allowing for the combination of evidence from different sources to arrive at degrees of belief.
- Unlike traditional methods, it combines evidence from multiple sources and assigns belief, plausibility and doubt to possible outcomes.

Need for DST

- **Combining Conflicting Evidence**: While Bayesian probability handles one piece of evidence at a time, it allows us to combine conflicting or incomplete evidence to offer a more holistic understanding.
- **Handling Uncertainty and Ignorance**: Bayesian models assume we have complete knowledge but DST can handle situations where we lack complete information, making it more flexible in uncertain environments.
- **Greater Flexibility**: Unlike Bayesian methods that work with fixed probabilities, it allows us to assess belief, giving us a clearer view of the range of possible outcomes.
- **Better Decision-Making**: It provides a structured approach to decision-making when faced with incomplete or uncertain information, helping us make more informed choices

Example

- In a room, four people A, B, C and D are present. The lights go out and when they come back on, B has been stabbed in the back. No one entered or left the room and B didn’t commit suicide. Our goal is to find who the murderer is.
- The set of possible hypotheses (Frame of Discernment) is:
- Either A or C or D has killed him.
- Either {A, C} or {C, D} or {A, D} have killed him.
- Or the three of them have killed him i.e {A, C, D}
- None of them have killed him {o} (let's say).

- The Power Set of P consists of all possible subsets of P including the empty set and the full set.
- For P = {(A, B, C, D)} , the power set 2^P is:
- $2^P = \{\emptyset, \{A\}, \{B\}, \{C\}, \{D\}, \{A, B\}\text{-----}\{B, C, D\}, \{A, B, C, D\}\}$

- Using the mass function, we calculate belief and plausibility for various hypotheses. For example, if we want to find the belief and plausibility for the hypothesis that {A, D} are involved, we proceed as follows:
- **Belief in K**: The belief for a hypothesis K (e.g {A, D}) is the sum of the masses of all subsets of K. If K= {A, D} then: Bel({A, D}) = m(A) + m(D) + m(A, D)
- **Plausibility in K**: Plausibility for K is the sum of masses for all sets that intersect with K. For K= {A, D} we would calculate: Pl({A, D}) = m(A) + m(D) + m(A, D) + m(A, B) + m(D, C) and so on for all subsets that intersect {A, D}.

- DST good at combining evidence from multiple sources. Suppose one witness claims A is guilty and another suggests that A and C are involved. It can combine these pieces of evidence into one belief:
- If the evidence for A is strong but unclear about whether C was involved, we combine the masses accordingly to get a unified belief and plausibility for different combinations of suspects.

Applications of Dempster Shafer Theory

- **Medical Diagnosis**: It helps combine multiple test results or symptoms to make a more accurate diagnosis by handling uncertainty in medical data.
- **Sensor Fusion**: It integrates data from different sensors to create a cohesive and accurate understanding even when individual sensors provide conflicting information.
- **Machine Learning**: It improves classification tasks by managing uncertainty in data allowing for more flexible and reliable predictions.
- **Robotics**: It is used in robotics to combine uncertain data from various sources, helping robots make better decisions in dynamic environments.

Challenges of Dempster Shafer Theory

- **Computational Complexity**: It can be resource-intensive especially with large sets of data or numerous evidence sources, making it harder to scale.
- **Conflict Handling:** It struggles with extreme conflicts between evidence sources which can lead to unreliable results if not properly managed.
- **Interpretability**: The results from DST may be difficult to interpret, particularly when combining evidence with high conflict or uncertainty.